



Interlaminar shear behavior of continuous and short carbon fiber reinforced polymer composites fabricated by additive manufacturing

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ABSTRACT

This study investigates the interlaminar shear behavior of 3D printed CFRP composites fabricated by fused deposition modeling (FDM) through a combined experimental and numerical study. Short beam shear (SBS) tests were performed to quantitatively characterize interlaminar shear strength (ILSS) of 3D printed continuous and short CFRP composites. Numerical simulations incorporating the extended finite element method (XFEM) and cohesive zone model (CZM) were conducted to rationalize the experiments. Two different combinations of continuous and short carbon fiber reinforcements were studied. The first composite layup made of 48 layers of continuous CFRP filaments was used to determine the intrinsic ILSS of the fully continuous CFRP composites. The second composite layup composed of 24 layers of continuous CFRP filaments and 24 layers of short CFRP filaments was used to determine the ILSS of the continuous and short CFRP composites. The average ILSS of these two composite layups were measured to be about 40.9 MPa and 24.4 MPa, respectively. Four distinct stacking sequences were implemented to examine the effect of stacking sequence of the continuous and short CFRP layers on the ILSS. The ILSS exhibited a significant sensitivity to the stacking sequence. The ILSS tends to increase with the number of the consecutive short CFRP layers. A failure mode transition from a brittle matrix cracking of short CFRP layers to cracking in the interface between the continuous and short CFRP layers was observed. The failure mode transition is attributed to the enhanced fracture strength and toughness of the short CFRP layers due to the increase in layer thickness.

1. Introduction

Carbon fiber reinforced polymer (CFRP) composites have been increasingly used as a structural material in the aerospace and automotive industries because of their superior mechanical and physical properties such as high specific stiffness and strength [1,2]. While various conventional manufacturing techniques such as filament winding, injection molding, automated tape laying and resin transfer molding have been successfully used for fabricating short or continuous CFRP composites, they typically involve a series of complex fabrication processes such as hand-layup, vacuum bagging, and high-temperature and pressure-assisted curing treatment (i.e., autoclaving) resulting in high production costs and low productivity. Although the injection molding is a rapid and low-cost manufacturing process among the traditional fabrication methods, it is only capable of fabricating short CFRP composite materials. In particular, the majority of the conventional manufacturing methods are not efficient in fabricating continuous CFRP composites with complex geometries as they often require expensive

metal molds. The costly and complex production processes for CFRP composites prevent CFRP composite materials from being widely used in the aerospace and automotive industries. Additive manufacturing, also known as 3D printing, has the potential to revolutionize the composite manufacturing industry because it enables geometric complexity, weight reduction, and fewer assemblies [3,4]. For example, 3D printing does not entail the abovementioned complex fabrication stages used in the traditional fabrication methods. In addition, 3D printing is able to fabricate CFRP composite parts with complex geometries as one-piece, which eliminates the requirement for the joining techniques such as adhesively bonded joints. While various 3D printing methods, including laminated object manufacturing (LOM), stereolithography (SLA), selective laser sintering (SLS), and fused deposition modeling (FDM), have been used to fabricate continuous or short CFRP composite materials [3, 4], only FDM with a dual extruder is capable of fabricating continuous and short CFRP composites simultaneously.

Although the tensile and flexural behaviors of 3D printed CFRP composites have been investigated extensively, little research has been

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reported on the interlaminar shear behavior of 3D printed continuous or short CFRP composites [5–15]. In layered composite materials, load transfer between adjacent layers is achieved by means of interlaminar shear and normal stresses. The interlaminar shear and normal strengths of composite materials have to be enhanced for an effective load transfer. Especially, interlaminar shear failure has been documented as one of the critical failure modes in CFRP composite materials under transverse loading [1,2]. Therefore, the interlaminar shear strength (ILSS) of CFRP composites plays a significant role in the structural integrity of the CFRP composite structures. The ILSS of traditionally manufactured CFRP composite materials has been reported for a wide range of fiber/matrix combinations through a well-established test method [2]. It was shown that the ILSS of CFRP composite materials is controlled by either the mechanical properties of the matrix material or the shear strength of fiber/matrix interface. In addition, the ILSS of CFRP composite materials exhibits a strong correlation with the void content. An increasing void volume fraction results in a significant degradation of the ILSS [2]. However, to the best of our knowledge, little research has been conducted to experimentally characterize the ILSS of 3D printed continuous and short CFRP composites [16,17]. For this reason, the interlaminar shear behavior of the 3D printed CFRP composites has to be properly addressed for the design and fabrication of structurally reliable 3D printed composite structures.

To fill this research gap, this study examines the interlaminar shear fracture behavior of 3D printed CFRP composites through a combined experimental and numerical study. Short beam shear (SBS) tests were performed to quantitatively characterize the ILSS of the 3D printed continuous and short CFRP composites. FDM was used to fabricate SBS specimens in two different combinations of continuous and short CFRP composite layup. The first composite layup is composed of 48 layers of continuous CFRP layers and used to measure the intrinsic ILSS of the 3D printed fully continuous CFRP composites. The second composite layup is composed of 24 continuous CFRP and 24 short CFRP layers in four

different stacking sequences to assess the effect of the varying stacking sequence on the ILSS of the 3D printed partially continuous and short CFRP composites. During the SBS tests, the digital image correlation (DIC) technique was used to measure the in-situ full field strain maps, which allowed us to observe interlaminar failure mechanisms. Moreover, numerical simulations incorporating the extended finite element method (XFEM) and cohesive zone model (CZM) were carried out to evaluate the effect of the stacking sequence of the continuous and short CFRP layers on the ILSS of the 3D printed partially continuous and short CFRP composites.

2. Experimental framework

2.1. Additive manufacturing and sample preparation

An FDM-based 3D printer with a dual extruder (Markforged Mark Two) was used to fabricate the SBS test samples for the ILSS tests. One extruder was used to print a continuous CFRP filament; the other was used to print a short CFRP filament. Fig. 1(a) illustrates a schematic of the FDM process of the continuous and short CFRP filaments, which also highlights a global printing coordinate (x, y, z) and a local material coordinate ($0^\circ, 90^\circ$) systems. The continuous CFRP filament used in this study is Markforged Carbon Fiber with a nominal diameter of 0.4 mm. An extrusion temperature of 232°C was used for printing continuous CFRP filaments. All the continuous fiber reinforced layers were aligned parallel to the y -direction, which is designated as 0° orientation in the local material coordinate system. The short CFRP filament used in this study is Markforged Onyx with a nominal diameter of 1.75 mm. Markforged Onyx is a short carbon fiber/Nylon 6 matrix filament. The extrusion temperature for short CFRP filaments was 275°C . A cross-ply $[0^\circ/90^\circ]$ raster pattern with 100% infill density was implemented in the short CFRP layers, which ensures in-plane quasi-isotropy in the short CFRP layers [2]. For both continuous and short CFRP layers, the layer

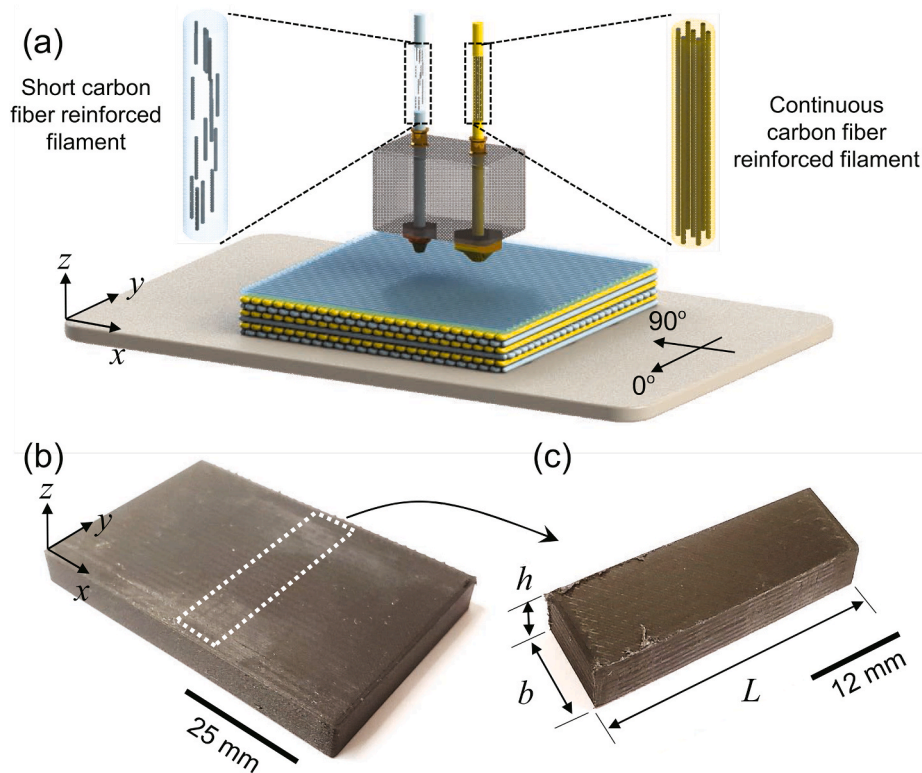


Fig. 1. (a) A schematic of FDM-based 3D printing of continuous and short carbon fiber reinforced filaments via a dual-extruder printer, (b) a 3D printed CFRP panel with a dimension of $40 \times 60 \times 6.25 \text{ mm}^3$, and (c) a SBS test sample with a dimension of $40 (L) \times 12 (b) \times 6.25 (h) \text{ mm}^3$, which was cut from the 3D panel by a saw cutter to eliminate boundary wall effects in the ILSS tests.

thickness was 125 μm . The SBS test samples were fabricated through two steps. First, small panels with a dimension of $40 \times 60 \times 6.25 \text{ mm}^3$ were fabricated via FDM. All the panels were printed in the xyz build orientation in accordance with the notation provided in ASTM 52921 [18]. Fig. 1(b) shows a 3D printed CFRP panel. Then, the 3D printed panels were cut into the SBS sample with a dimension of $40 \times 12 \times 6.25 \text{ mm}^3$ using a saw cutter. Fig. 1(c) shows an SBS test sample after the cutting process. The second step aims to eliminate the edge effects arising from the inconsistent fiber content or alignment and the wall structure within regions close to the panel edges. The dimensions of the SBS specimens were carefully adjusted to meet the recommended dimensions provided in the standard testing procedure ASTM D2344 [19].

Two different combinations of continuous and short carbon fiber reinforcements were studied to characterize the interlaminar shear behavior of the 3D printed CFRP composite with full and partial continuous fiber reinforcements. To illustrate the layout of the 3D printed CFRP composite samples, a short-hand notation was implemented based on the well-established notation for the composite laminates with minor alterations. As an example, in $[(CF)_i, (SF)_j]_{NS}$, CF and SF inside the parentheses denote the continuous and short CFRP layers, respectively. The subscription i denotes the number of consecutive continuous or short CFRP layers. The fiber orientation information was not included in the notation as the continuous carbon fibers were aligned with the 0° direction in the continuous CFRP layers, while the short CFRP layers were fabricated with a cross-ply $[0^\circ/90^\circ]$ layout. The subscriptions N and S denote the number of repetitions of the selected stacking sequence and the symmetry of the layout, respectively. As recommended in the standard procedure of the SBS test [19], all the SBS test samples were fabricated with a symmetric stacking sequence around the mid-plane. Accordingly, the first sample layout $[(CF)]_{48S}$ was composed of 48 continuous CFRP layers as shown in Fig. 2(a). This sample was used to determine the intrinsic ILSS of the 3D printed fully continuous CFRP composites. The second configuration was composed of 24 continuous CFRP and 24 short CFRP layers in four distinct stacking sequences including $[(CF)_1, (SF)_1]_{24S}$, $[(CF)_2, (SF)_2]_{12S}$, $[(CF)_3, (SF)_3]_{8S}$, and $[(CF)_4, (SF)_4]_{6S}$ as shown in Fig. 2(b–e). These samples were examined to assess the ILSS of the 3D printed partially continuous and short CFRP composites as well as the effect of the stacking sequence on the ILSS. Fig. 2(b–e) show the optical microscopy images of the corresponding side view of each stacking sequence, where the light-gray and dark-gray shaded regions denote continuous and short CFRP layers, respectively.

2.2. Short beam shear test

The short beam shear (SBS) tests were performed in accordance with ASTM D2344 [19] to determine the ILSS of the 3D printed CFRP composite samples. The SBS test samples were loaded in a 3-point bend load configuration as shown in Fig. 3(a). The displacement was applied at the center of the samples through the loading pin, while the samples were held by the side support pins. The distance between the support pins was adjusted to be 4-fold of the sample thickness (i.e., $S = 4h$). A special care was taken to maintain the alignment of the samples such that their longitudinal axis is perpendicular to the loading and support pins. All the tests were carried out under displacement control using the MTS Insight 5 kN computer controlled universal testing frame at room temperature. A ramp displacement profile with a constant crosshead movement rate of 1 mm/s was applied. The applied cross-head displacement, force, and in-situ digital images of the sample surface were recorded simultaneously. These tests were repeated on at least three samples for each configuration to obtain the statistical distribution of the measured ILSS.

The in-situ optical images of the sample side surface were recorded using a high-resolution digital monochrome camera of 2448×2048 pixels (FILR Systems, Chameleon CM3-U3-50S5M – CS USB3) with a 36 mm lens (Edmund Optics). An image acquisition rate of 1 Hz was used to capture the images of the interlaminar damage nucleation and evolution. The field of view of the images was adjusted to be about $42 \times 35 \text{ mm}^2$ to capture the entire span of the samples. Approximately 300 sequential optical images with a resolution of about $17\mu\text{m}/\text{pixel}$ were captured for each sample. A white colore spray painter was used to introduce a random speckle pattern on the side surface of the samples prior to the tests, which facilitated full field strain measurements by DIC. Fig. 3(b) shows the randomly introduced speckle pattern with an average speckle size of about $50 \mu\text{m}$. The collected images were analyzed using 2D DIC software Ncorr [20] to measure the full field strain distribution on the side surfaces of the samples. The window size of the DIC analysis was selected as 10×10 pixels, which ensures existence of at least 3 speckle features within the selected window size. The in-situ full field strain maps were used to understand the interlaminar damage nucleation and propagation behavior of the examined CFRP composite samples.

The ILSS τ_{ILSS} of the 3D printed continuous and short CFRP composites were determined from the measured force and displacement data. Following the standard data reduction scheme provided in the ASTM D2344, for a sufficiently small support span length-to-sample thickness ratio, the ILSS can be estimated using the definition of

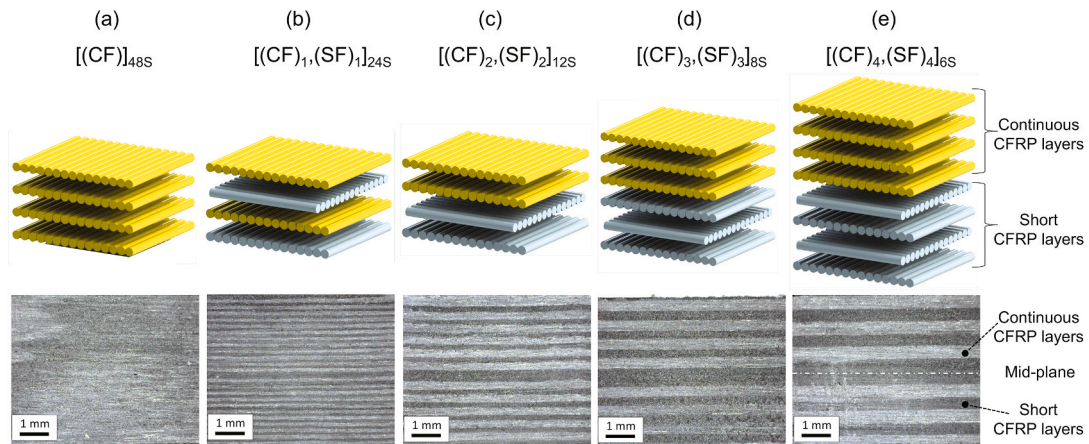


Fig. 2. Schematics of the 3D printed CFRP composite samples showing the layout and stacking sequences of the short and continuous CFRP layers for (a) the fully continuous CFRP composite layout $[(CF)]_{48S}$ and (b–e) partially continuous and short fiber CFRP composite layouts with variable stacking sequences (b) $[(CF)_1, (SF)_1]_{24S}$, (c) $[(CF)_2, (SF)_2]_{12S}$, (d) $[(CF)_3, (SF)_3]_{8S}$, and (e) $[(CF)_4, (SF)_4]_{6S}$. The optical microscopy images presented in the second row show the corresponding side view of the samples, where the light-gray shaded regions denote continuous CFRP layers, while the dark-gray shaded regions denote short CFRP layers.

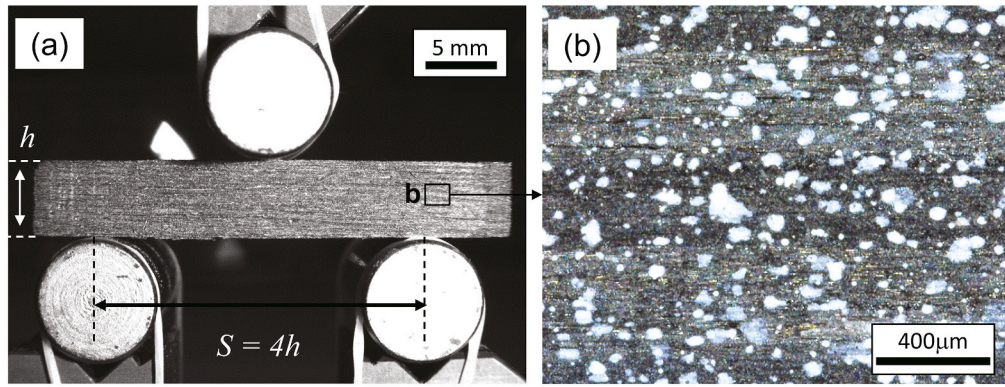


Fig. 3. (a) Short beam shear test configuration for ILSS characterization of the examined 3D printed CFRP composite, (b) a close-up optical microscope image of the sample side surface showing the randomly introduced speckle pattern with an average speckle size of 50 μm for DIC strain measurement.

maximum shear stress as

$$\tau_{ILSS} = \tau_{max} = \frac{3}{4} \frac{P_{max}}{b h} \quad (1)$$

where P_{max} is the maximum force measured in the test, b is the sample width, and h is the sample thickness.

2.3. Microstructural characterization

The microstructure of the continuous and short CFRP filaments and 3D printed CFRP composite samples were examined using an optical microscope (Leica DM4 B) and scanning electron microscope (SEM) (Zeiss ULTRA-55 FEG SEM). The short and continuous CFRP filaments were examined to gather information about the fiber geometry and volume fraction, while the cross-section of the 3D printed samples was inspected to characterize the manufacturing-induced defects. The CFRP filaments and 3D printed samples for microstructural characterization were first mounted in epoxy using a low viscosity resin and hardener system (Buehler Epoxycure). Then, the cross-section of the samples was polished with 400, 600, 800, and 1200-grit grinding papers using a polisher (Buehler VibroMet 2 Vibratory Polisher). The polished sample surfaces were coated with a conductive thin layer of gold using the sputter coating technique (Ted Pella 108 Auto/SE Sputter Coater). The optical and SEM microscope images were analyzed using an image processing tool to quantitatively evaluate the microstructural features such as fiber and void volume fractions of the 3D printed samples. In addition, the void volume fractions of the 3D printed samples were verified by theoretical and experimental density estimations using the rule of mixtures approach.

3. Finite element model

Finite element (FE) simulations employing the extended finite element method (XFEM) in conjunction with the cohesive zone model (CZM) were performed to understand the experimentally observed interlaminar shear behavior of the 3D printed continuous and short CFRP composites. The XFEM introduced by Blytschko and Black [21] is a powerful numerical modeling technique for simulating nucleation and growth of various discontinuities such as bulk cracks without remeshing of the FEM model. The XFEM with a proper damage model has been successfully employed for modeling matrix-cracking failure of CFRP composites [22–26]. In addition, the CZM is one of the most widely used techniques to predict the interfacial fracture behavior of CFRP composite materials [27–31]. Both XFEM and CZM have been demonstrated to be effective in modeling fracture behavior of 3D printed materials [32,33]. In this study, the XFEM was used to simulate the matrix cracking within the 3D printed short CFRP layers as the short CFRP layers experimentally revealed matrix cracks arising from the failure of

short fiber-matrix interfaces. The consecutive short CFRP layers were effectively modeled as one part with XFEM mesh using maximum nominal stress damage nucleation criterion. On the other hand, the CZM was utilized to model the interfacial crack nucleation and growth on the interfaces between short and continuous CFRP layers. The experimental results showed that the interlaminar shear failure takes place through brittle cracking of the short CFRP layers or failure of the interface between the continuous and short CFRP layers. The continuous fiber-/matrix interfaces within the continuous CFRP layers were observed to be intact in most of the experiments. Therefore, based on the experimental observations, the failure of the continuous CFRP layers was not considered in the numerical models.

The FE models were developed in a commercial FE software ABAQUS 6.13 [34]. The SBS sample was modeled with 4-node continuum plane strain quadrilateral elements with reduced integration and hourglass control (CPE4R). The continuous CFRP layers were effectively modeled as an isotropic elastic material with Young's modulus $E_{CF} = 50 \text{ GPa}$ and Poisson's ratio $\nu_{CF} = 0.28$ [35]. An elastic-plastic constitutive model was employed to model the mechanical behavior of the short CFRP layers. The mechanical properties of the short CFRP layers were independently measured from tensile tests such as Young's modulus $E_{SF} = 1.6 \text{ GPa}$, Poisson's ratio $\nu_{SF} = 0.28$, and yield strength $\sigma_y|_{SF} = 64.2 \text{ MPa}$. The yield strength was governed by a Von-mises type isotropic plasticity model. A modest strain hardening exponent was implemented as seen in the tensile tests. The loading and support pins were modeled as analytically rigid for the sake of computational efficiency. The shear and normal interactions between the SBS sample and loading/support pins were simulated as a frictionless-hard contact interaction. As shown in Fig. 4(a), the geometry and boundary conditions were selected to be identical to the experimental counterparts. A vertical displacement input was applied at the center of the loading pin, while the centers of the support pins were restricted along the vertical and horizontal directions. The rotation was allowed at the centers of the loading/support pins.

Fig. 4(b) shows a close-up view of the FE model showing three consecutive (i.e., continuous-short-continuous CFRP) layers. An XFEM mesh was implemented within the short CFRP layers, while cohesive surfaces were employed between the continuous and short CFRP layers. A maximum nominal stress criterion was used for the damage initiation in the XFEM elements because of the orthotropic damage parameters arising from the layered nature of the 3D printed CFRP samples. The short CFRP layers were found to be weaker in the out-of-plane directions relative to the in-plane directions [2]. For the damage evolution of the bulk cracks within the short CFRP layers, an energy-based damage evolution criterion was used. Accordingly, the critical shear stress $\hat{\tau} = 61 \text{ MPa}$ and fracture energy $\Gamma_0 = 3100 \text{ J/m}^2$ were used as the bulk fracture parameters for the XFEM elements. The bulk fracture parameters were evaluated by the compact tension fracture test. The

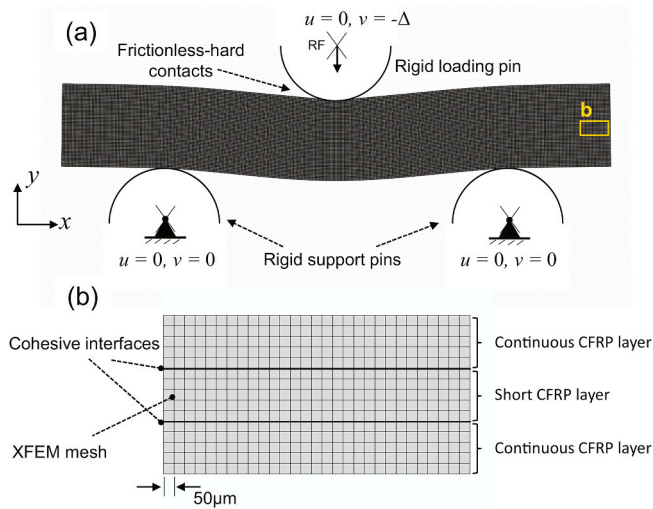


Fig. 4. (a) The employed finite element model showing the SBS test sample geometry and applied boundary conditions in the FE numerical simulations, (b) a close-up view of the region b in (a) showing the details of the XFEM mesh within the short CFRP layers and cohesive surfaces between the continuous and short CFRP layers.

cohesive surfaces were governed by a bilinear cohesive traction-separation [31]. For the interfacial damage nucleation, a maximum nominal stress criterion was used. An energy-based damage evolution criterion was used for the interfacial damage propagation. The three independent parameters of the bilinear cohesive traction-separation curve were taken as the initial slope of the curve $K = 10^6 \text{ MPa/mm}$, the critical cohesive shear stress $\hat{\tau} = 60 \text{ MPa}$, and the cohesive fracture energy $\Gamma_0 = 1100 \text{ J/m}^2$. The $\hat{\tau}$ and Γ_0 of the bilinear cohesive traction-separation curve were iteratively calibrated to match the experimental data from the SBS test. The fracture mode-mixity was not considered as the SBS test geometry leads to nearly pure shear fracture conditions. The global mesh size was adjusted as 50 μm after a convergence study verifying the mesh-independency of the model. It was also confirmed that the mesh size was sufficiently smaller than the

cohesive zone length at the interface [36,37].

4. Results and discussions

4.1. Microstructure characterization

The microstructural characterization of continuous and short CFRP filaments is essential to examine the fiber geometry, alignment and volume fraction. All these physical properties of the filaments affect the mechanical properties of the 3D printed CFRP materials. Therefore, the continuous and short fiber reinforced filaments used in this study were examined through optical and SEM cross-section microscopy. Fig. 5(a–c) show the cross-sectional optical microscope images of the continuous carbon fiber filament. The entire filament with a nominal diameter of about 0.4 mm is depicted in Fig. 5(a). The light gray-shaded small circular regions denote the continuous carbon fibers, while the dark gray-shaded region within the filament is the matrix material (sizing agent) holding the fibers together. Fig. 5(b) shows a close-up view of the area b shown in Fig. 5(a), where one can recognize a few closely packed bundles of continuous carbon fibers. A higher magnification image of one of the fiber bundles shown in Fig. 5(b) is shown in Fig. 5(c), revealing an average continuous fiber diameter of about 8 μm . Moreover, Fig. 5(d–f) show the cross-sectional SEM images of the short carbon fiber filament. Fig. 5(d) shows the entire filament with a nominal diameter of about 1.75 mm. Fig. 5(e) shows a close-up view of the area e shown in Fig. 5(d), which is nearly located at the center of the filament and shows a random distribution of the short fibers. Fig. 5(e) suggests that short fibers are aligned parallel with the longitudinal axis of the filament. A single short carbon fiber was captured in Fig. 5(f), showing an average short fiber diameter of about 6 μm . The average short carbon fiber length was about 60 μm from the SEM inspection of 3D printed CFRP samples. The average continuous and short carbon fiber volume fractions of the filaments were calculated to be $24.8 \pm 0.6\%$ and $22 \pm 1\%$ through digital image analyses of the microscope images, respectively.

In addition, a proper characterization of the manufacturing-induced defects in the 3D printed CFRP composites is crucial for understanding the mechanical behavior of the 3D printed CFRP composite materials. Accordingly, cross-sectional SEM inspections were performed to evaluate the microstructures of the 3D printed continuous and short CFRP

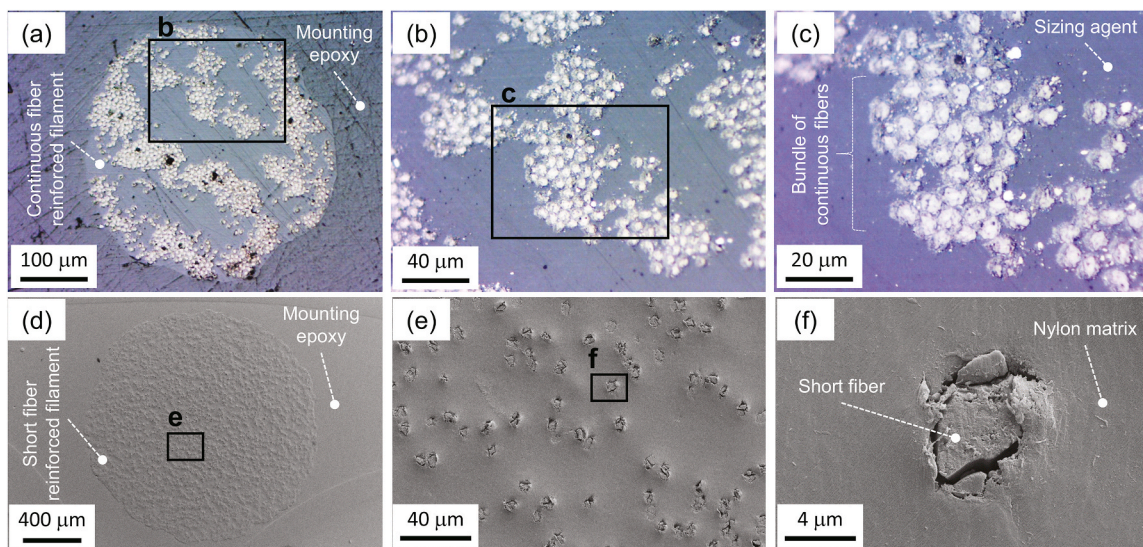


Fig. 5. Combined optical microscope and SEM images of (a–c) continuous and (d–f) short CFRP filaments. (a) An optical microscope image of the continuous CFRP filament showing the entire filament with a nominal diameter of 0.4 mm, (b) a close-up view of the region b in (a) showing closely packed bundles of continuous carbon fibers, (c) a close-up view of the region c in (b) revealing an average continuous carbon fiber diameter of about 8 μm , (d) an SEM image of the entire filament with a nominal diameter of 1.75 mm, (e) a close-up view of the region e in (d) showing the uniform distribution of the short fibers, (f) a close-up view of the region f in (e) showing a single short carbon fiber with an average diameter of about 6 μm .

layers as shown in Fig. 6(a). Fig. 6(b) and (c) show the cross-section SEM images of the continuous CFRP layers at two different magnifications. Fig. 6(b) shows the microstructure of approximately four consecutive unidirectional continuous CFRP layers, in which one can identify several micro-voids located at the layer boundaries. A close-up view of the region c shown in Fig. 6(b) is presented in Fig. 6(c), which highlights a few interfacial micro-cracks at the continuous fiber-matrix interfaces. Such micro-cracking can be associated with the surface damage induced by the sample cross-sectioning process. Moreover, Fig. 6(d) and (e) show the cross-section SEM images of the four adjacent short CFRP layers at two different magnifications. Fig. 6(d) shows two distinct types of voids within a 3D printed short CFRP layer: (1) inter-beads voids introduced during the 3D printing process and (2) micro-voids existing within the filaments arising from the fiber pull-out due to the brutal shear force in the polishing process. Fig. 6(e) shows a close-up view of the region e marked in Fig. 6(d), depicting two distinct features: (1) an inter-layer void and (2) a couple of interfacial micro-cracks at the short fiber-matrix interfaces. Similarly, the former can be related with the 3D printing process and the latter can be attributed to the surface damage due to the sample cross-sectioning process. The SEM images were processed by a digital image processing tool and the void volume fractions were estimated for each layup. The void content was also verified by physical density measurements on the samples. The average values of

continuous and short carbon fiber volume fractions as well as the void volume fraction of each layup are provided in Table 1. The continuous CFRP samples revealed about 3–4% void volume fraction. The predicted void volume fraction in this study was found to be relatively smaller compared to the reported value of about 10% in Ref. [6,38,39]. This discrepancy can be attributed to the elimination of edge voids in this

Table 1

Continuous and short fiber volume fractions, void volume fractions, and ILSS of the examined 3D printed composites in different layups.

Layup	Continuous fiber volume fractions	Short fiber volume fractions	Void volume fraction (%)	ILSS (MPa)
$[(CF)]_{48S}$	24.1	0	3.4	40.90 ± 3.06
$[(CF)_1, (SF)_1]_{24S}$	12.0	9.5	8.7	22.90 ± 1.35
$[(CF)_2, (SF)_2]_{12S}$	11.9	9.6	8.8	23.81 ± 0.55
$[(CF)_3, (SF)_3]_{8S}$	12.1	9.5	8.5	23.05 ± 1.32
$[(CF)_4, (SF)_4]_{6S}$	12.1	9.5	8.5	28.10 ± 0.80

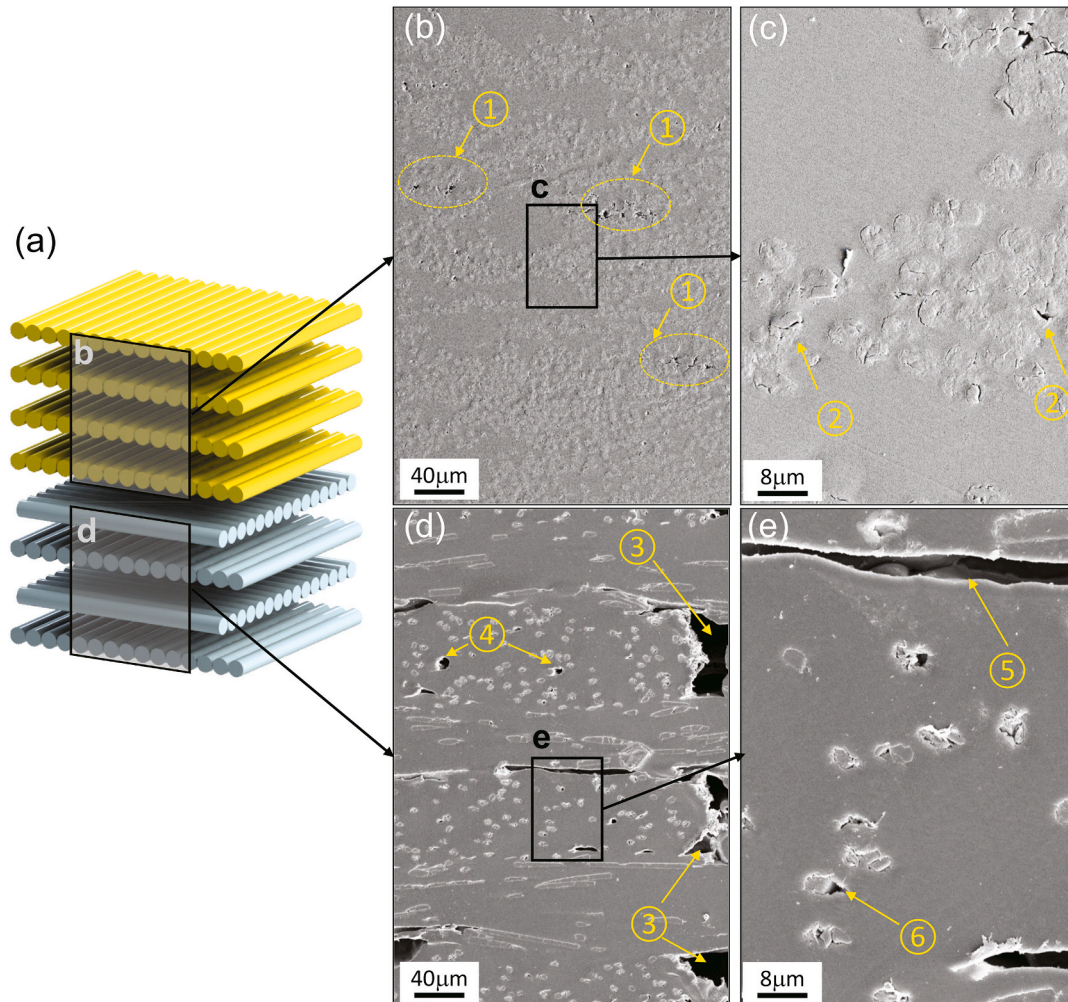


Fig. 6. (a) A 3D schematic showing the stacking sequence of continuous and short CFRP layers of the SEM cross-section sample, (b) a cross-sectional SEM image of the 3D printed continuous CFRP layers revealing micro-voids at the layer boundaries, (c) a close-up view of the region c shown in (b) depicting ② interfacial micro-cracks at the continuous fiber-matrix interfaces, (d) a cross-sectional SEM image of the 3D printed short CFRP layers revealing two types of voids: ③ large inter-beads voids and ④ micro-voids within the printed filaments, (e) a close-up view of the region e shown in (d) depicting ⑤ an interlayer void and ⑥ interfacial micro-cracks at the short fiber-matrix interfaces.

study. As a result of the nature of 3D printing, large voids tend to form within the edges of the 3D printed samples due to the U-turn motion of the printing nozzle [38]. Such large voids were excluded in this study as the examined samples were cut from the 3D printed panels. On the other hand, the void volume fraction of the continuous and short CFRP samples were about 8–10%. Noteworthy is that the void volume fraction was considerably insensitive to the stacking sequence of the continuous and short CFRP layers. In this case, the void volume fractions of the continuous and short CFRP samples were found to be close to the lower bound of the typical void volume fraction range of 9–15% reported in Ref. [4,40,41].

4.2. Interlaminar shear behavior of 3D printed fully continuous CFRP composites

The interlaminar shear behavior of the 3D printed fully continuous CFRP composites with $[(CF)]_{48S}$ layup was characterized through the SBS tests. Fig. 7 shows the maximum shear stress-displacement curves obtained from testing of three different samples. The maximum shear stress, instead of force, was used to present the experimental data, which eliminates the sample-to-sample variations in the measured force due to the minor variation of sample width. The maximum shear stress was calculated by normalizing the measured force at each level with the sample thickness and width given in Eq. (1). Therefore, the maximum values of the maximum shear stress plotted in Fig. 7 correspond to the ILSS. As shown in Fig. 7, the maximum shear stress-displacement curves initially exhibit a linear deformation with a slight non-linearity within the very beginning of the curves. This initial non-linearity is attributed to the establishment of the initial contact between the samples and the loading and support pins. The maximum shear stress-displacement curves also show a slight nonlinearity prior to their maximum values are attained. The presence of such non-linearity is associated with the interlaminar damage nucleation. Within the non-linear portions of the curves, the maximum shear stress-displacement curves exhibit a monotonic increase until a critical stress state, at which the interlaminar damage reaches a critical level. Thereafter, the maximum shear stress-displacement curves exhibit a gradual decreasing trend associated with the interlaminar damage evolution. The ILSS of the 3D printed fully continuous CFRP composites was recorded as 40.9 ± 3.06 MPa, which

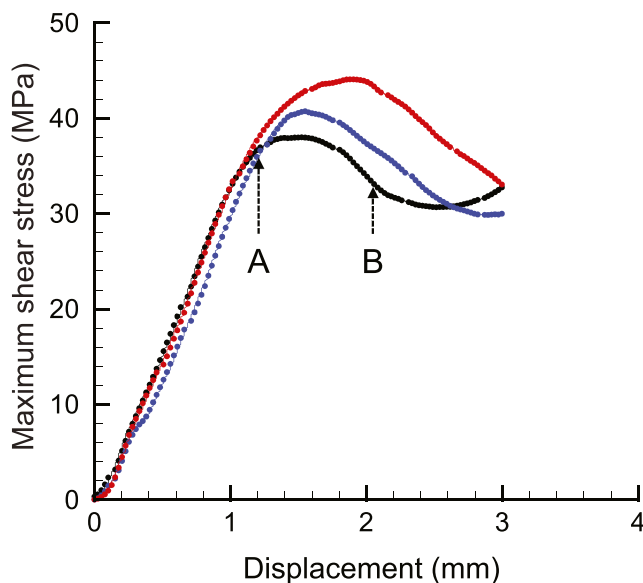


Fig. 7. A representative set of maximum shear stress – displacement curves obtained from the SBS tests of the 3D-printed fully continuous CFRP composites, $[(CF)]_{48S}$. The points A and B marked on one of the curves denote the interlaminar damage nucleation and propagation stages, respectively.

was found to be 30% greater than the ILSS reported by Ref. [16,17] for the same material system. This discrepancy can be attributed to the difference in the void fractions due to the variation of the additive manufacturing process parameters or the edge defects generated by the sample preparation process such as cutting.

Fig. 8(a–d) show the shear strain ϵ_{xy} and vertical strain ϵ_{yy} distributions evaluated by the DIC method at points A and B marked in Fig. 7 to examine strain states corresponding to the interlaminar damage nucleation and propagation phases, respectively. One can note that the shear strain field reveals an unsymmetrical distribution between the left- and right-hand sides of the sample. Such an unsymmetrical distribution can arise from the off-centered loading pin and/or the realignment of the sample with the damage nucleation. The former is eliminated as a special care was taken for the proper initial alignment of the sample with respect to the loading fixture. In the nucleation stage at point A, Fig. 8(a) reveals a uniformly distributed shear strain through the thickness on the right-hand side of the sample. However, the vertical strain component shows a few localized strain spots with a positive vertical strain. These localized strain sites are the locations where the opening occurs between the layers and the interlaminar damage nucleates. In the damage propagation stage at point B, the shear strain continues increasing uniformly as shown in Fig. 8(c). The vertical strain shows the damage propagation more vividly as the localized strain sites become larger in size as shown in Fig. 8(d). To further examine the strain distribution through the sample thickness, Fig. 8(e) shows the shear strain ϵ_{xy} and vertical strain ϵ_{yy} distributions along the vertical line cut shown in Fig. 8(a) and (b) as a function of the distance from the top surface. The shear strain shows a parabolic distribution with a maximum value around the mid-plane, while four spikes can be identified in the vertical strain distribution corresponding to the layers of the interlaminar damage nucleation. This observation is consistent with the interlaminar cracking pattern captured on the side surface of the sample at the end of the tests as shown in Fig. 8(f). The interlaminar cracks can be located above and below the mid-plane. The cracks above the mid-plane show larger opening as consistent with the vertical strain distribution presented in Fig. 8(e). In addition, a close-up view of the region g marked in Fig. 8(f) shows a rough interlaminar crack growth path. Such a large roughness on the crack surfaces can be attributed to the frictional-sliding on the crack surfaces during the damage evolution.

4.3. Interlaminar shear behavior of 3D printed partially continuous and short CFRP composites

The interlaminar shear behavior of the 3D printed partially continuous and short CFRP composites in the four different stacking sequences was examined by the SBS test. Fig. 9 shows the maximum shear stress-displacement curves obtained from testing of the four different stacking sequences. Three maximum shear stress-displacement curves are presented for each stacking sequence to show the sample-to-sample variations. Similar to the fully continuous CFRP composite samples, the maximum shear stress-displacement curves of the partially continuous and short CFRP composites in all stacking sequences exhibit a linear deformation within the initial portions of the curves. The initial slopes of the curves were found to be insensitive to the stacking sequence, which suggests that the variation of the stacking sequence does not result in a significant variation in the flexural stiffness of the SBS samples. However, one can note two major differences in the maximum shear stress-displacement curves with the varying stacking sequence. First, the SBS samples with the layup $[(CF)_1, (SF)_1]_{24S}$ and $[(CF)_2, (SF)_2]_{12S}$ exhibit a subtle non-linearity prior to the ILSS, while the samples with $[(CF)_3, (SF)_3]_{8S}$ and $[(CF)_4, (SF)_4]_{6S}$ layups show a considerable nonlinearity before reaching the ILSS. The substantial nonlinearity observed in the latter case can be attributed to the enhanced plastic deformation within the short CFRP layers with the increasing number of consecutive short CFRP layers. Second, the maximum shear

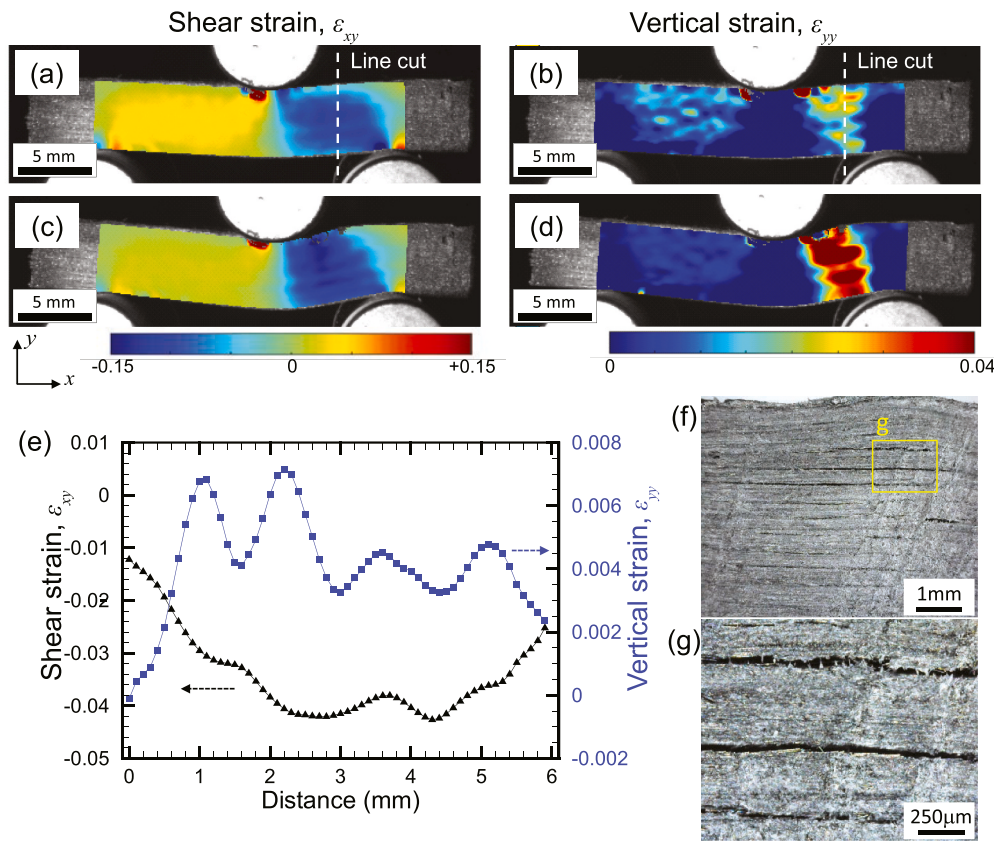


Fig. 8. The shear strain ϵ_{xy} and vertical strain ϵ_{yy} fields evaluated by DIC showing (a, b) the damage nucleation state at point A marked in Fig. 7 and (c, d) propagated damage state at point B marked in Fig. 7, (e) the shear strain ϵ_{xy} and vertical strain ϵ_{yy} distributions along the vertical line cut shown in (a) and (b) as a function of the distance from the top surface, (f) an optical image showing the corresponding interlaminar cracking pattern on the sample side surface after completion of the test, (g) a close-up view of the region g marked in (f) depicting interlaminar cracks with rough crack surfaces.

stress-displacement curves of the stacking sequences $[(CF)_1, (SF)_1]_{24S}$ and $[(CF)_2, (SF)_2]_{12S}$ show intermittent drops in the maximum stress after the point of damage nucleation. This is a characteristic of the unstable interlaminar damage nucleation and propagation at each maximum shear stress drop. The maximum shear stress-displacement curves of the stacking sequence $[(CF)_3, (SF)_3]_{8S}$ still show similar stress drops, which become less remarkable. However, the maximum shear stress-displacement curves of the stacking sequence $[(CF)_4, (SF)_4]_{6S}$ exhibit a continuous gradual decaying trend in the maximum shear stress after the ILSS is reached. Such a continuous decaying trend can be associated with the stable interlaminar damage nucleation and evolution. The average ILSS of the 3D printed partially continuous and short CFRP composites was found to be about 24.4 ± 2.4 MPa. However, we have identified a strong correlation between the ILSS and the stacking sequence of the continuous and short CFRP layers, which will be discussed in Section 4.4 thoroughly.

To further understand these two distinct types of interlaminar damage nucleation and propagation, the shear strain ϵ_{xy} and vertical strain ϵ_{yy} distributions at the different stages of the damage evolution along the optical images of the interlaminar cracking morphology were examined for the SBS test samples with $[(CF)_2, (SF)_2]_{12S}$ and $[(CF)_4, (SF)_4]_{6S}$ layouts, respectively. First, Fig. 10(a–f) show the shear strain ϵ_{xy} and vertical strain ϵ_{yy} distributions evaluated by the DIC method at points A, B, and C marked in Fig. 9(b). These three points correspond to three consecutive shear stress drops recorded in the maximum shear stress-displacement curve. At the first shear stress drop (point A), Fig. 10(a) and (b) reveal the strain localizations at two different layers above and below the mid-plane of the sample. Then, at the second shear stress drop (point B), one more layer with a strain localization appears between the initially damaged layers as shown in Fig. 10(c) and (d). Finally, at the third shear stress drop (point C), these strain localizations become denser and spread to all the layers as shown in Fig. 10(e) and (f). Noteworthy is that both shear and vertical strain

components shown in Fig. 10(g) reveal the exact locations of the initially damaged layers approximately 1.4 mm and 0.6 mm apart from the mid-plane of the sample. These damaged layers can also be identified through the optical image showing the interlaminar cracking pattern captured on the side surface of the sample at the end of the test as shown in Fig. 10(h). Here, one can note that the interlaminar cracks are concentrated above the mid-plane in consistent with the in-situ strain measurements by DIC. More importantly, Fig. 10(i), a close-up view of the region i shown in Fig. 10(h), clearly reveals that the interlaminar cracks nucleate and propagate within the short CFRP layers.

Second, Fig. 11(a–f) show the shear ϵ_{xy} and vertical strain ϵ_{yy} distributions measured by the DIC method at points A, B, and C marked in Fig. 9(d). In this case, the strain state at point A corresponds to the damage nucleation stage, at which the nonlinearity commences in the maximum shear stress-displacement curve, while the strain states at points B and C represent the damage evolution stages in the maximum and decaying parts of the maximum shear stress-displacement curve. In the damage nucleation stage at point A, Fig. 11(a) shows a symmetric and uniform shear strain distribution around the central load line. This suggests that the sample undergoes a uniform deformation prior to the damage. Inspecting the vertical strain component shown in Fig. 11(b), one can identify a few localized strain sites with a positive vertical strain through the sample thickness. These locations can be perceived as the crack nucleation sites. In the damage propagation stage at point B, the shear strain distribution shown in Fig. 11(c) exhibits strain localizations at four different layers, in which the interlaminar crack growth takes place. However, the vertical strain distribution shown in Fig. 11(d) exhibits only two strain concentrations with a large positive strain. This implies that the damage evolution occurs nonuniformly. In the later stage of the damage evolution at point C, the interlaminar cracks propagate in the same layers with the fully developed shear and vertical strain fields as can be seen in Fig. 11(e) and (f). The line distributions of the ϵ_{xy} and ϵ_{yy} strain components shown in Fig. 11(g) shed light on the

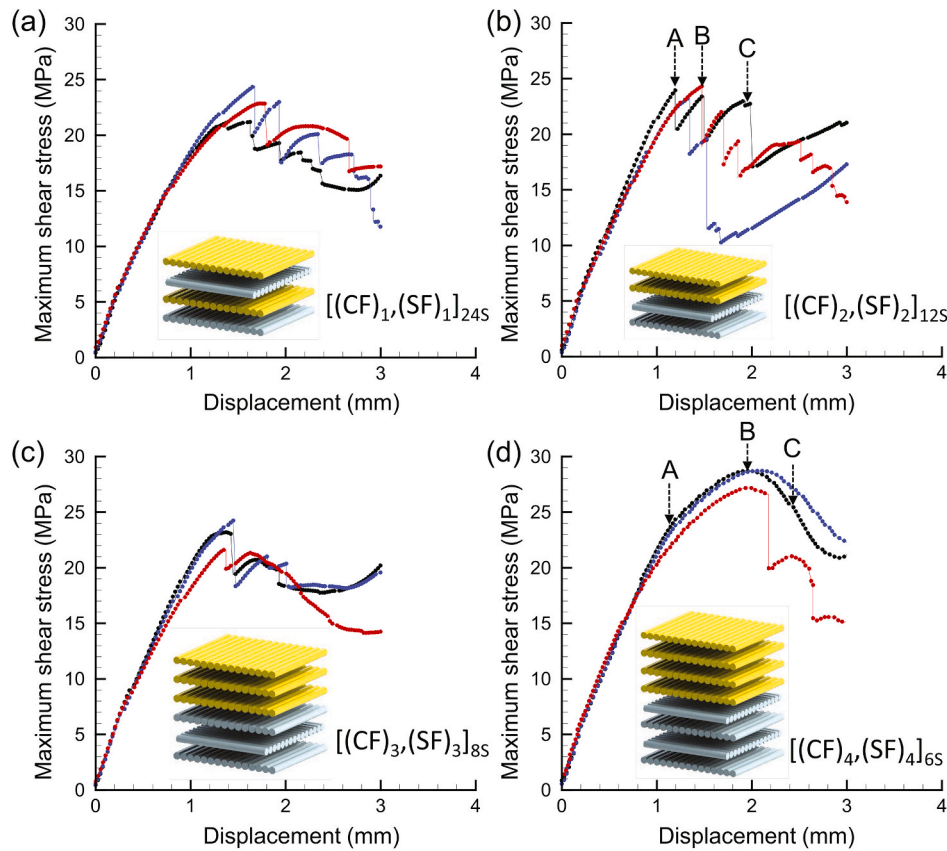


Fig. 9. Maximum shear stress-displacement curves obtained from the SBS tests of the 3D-printed partially continuous and short CFRP composite samples with the stacking sequences of (a) $[(CF)_1, (SF)_1]_{24S}$, (b) $[(CF)_2, (SF)_2]_{12S}$, (c) $[(CF)_3, (SF)_3]_{8S}$, and (d) $[(CF)_4, (SF)_4]_{6S}$. The points A, B, and C marked on (b) and (d) denote different stages of damage evolution.

interlaminar damage nucleation. The shear strain component displays a parabolic distribution with a maximum around the mid-plane. The vertical strain component shows positive strain spikes at four different locations, which indicates the exact locations of the interlaminar damage nucleation. These locations are consistent with the interlaminar cracking pattern captured at the end of the test as shown in Fig. 11(h). One can locate these four delaminated layers almost symmetrically positioned above and below the mid-plane. More interestingly, all cracks nucleate and propagate at the interface of the continuous and short CFRP layers. A close-up view of the region i marked in Fig. 11(h) shows the interfacial cracks more clearly. It should be noted that the varying stacking sequence from $[(CF)_2, (SF)_2]_{12S}$ to $[(CF)_4, (SF)_4]_{6S}$ resulted in a failure mode transition from matrix-cracks to interfacial cracks.

4.4. Effect of stacking sequence of continuous and short CFRP layers on ILSS

Fig. 12 shows the variation of the ILSS with the number of the consecutive short CFRP layers. We observed a monotonic increasing trend with the increasing number of the consecutive short CFRP layers. This increasing trend was accompanied by a failure mode transition from the brittle matrix cracking of the short CFRP layers to the failure of the interface between the continuous and short CFRP layers. This monotonic increasing trend can be attributed to the enhanced bulk fracture strength and toughness of the short CFRP layers with the increasing thickness. In fact, a positive correlation between the layer thickness of compliant adhesives between two rigid substrates and their effective fracture strength and toughness was previously reported. It was shown that an increasing layer thickness enhances the plastic work dissipation within a compliant adhesive layer, which in turn leads to

effectively improved strength and fracture toughness [42–45]. Similarly, it can be hypothesized that the total thickness of the short CFRP layers between the stiff continuous CFRP affects the ILSS by controlling the level of the plastic deformation within the short CFRP layers. The experimental observations strongly support this hypothesis. For example, the stacking sequence of $[(CF)_2, (SF)_2]_{12S}$ exhibited almost no plasticity with brittle matrix cracking in the short CFRP layers due to the thin layer thickness. However, the stacking sequence of $[(CF)_4, (SF)_4]_{6S}$ revealed a significant plasticity within the short CFRP layers due to effectively improved fracture strength and toughness of the short CFRP layer as a result of the greater layer thickness.

To further understand the effect of the stacking sequence on the ILSS and the failure mode, the FE models of the $[(CF)_2, (SF)_2]_{12S}$ and $[(CF)_4, (SF)_4]_{6S}$ were created. Both XFEM and CZM methods were used to simulate the experimentally observed failure modes, i.e. brittle matrix cracking and interfacial cracking, respectively. The bulk fracture parameters of the XFEM elements were reduced by a factor of 2 to mimic the reduced fracture strength and toughness due to the smaller layer thickness of the short CFRP in the simulations of the stacking sequence of $[(CF)_2, (SF)_2]_{12S}$. Fig. 13 shows the numerically derived maximum shear stress-displacement curve for the stacking sequence of $[(CF)_2, (SF)_2]_{12S}$, along with one of the experimentally measured counterparts. The inset snapshots in Fig. 13 reveal three subsequent stages of the damage evolution at points A, B and C marked in the maximum shear stress-displacement curve. Here, the color bar represents the status of the damaged XFEM elements ranging from 0 to 1, where 0 and 1 denote no damage and full damage states, respectively. The numerically obtained maximum shear stress-displacement curve shows a linear deformation within the initial part of the curve, similar to the experimental counterpart. At a critical stress level (point A), a nonlinearity

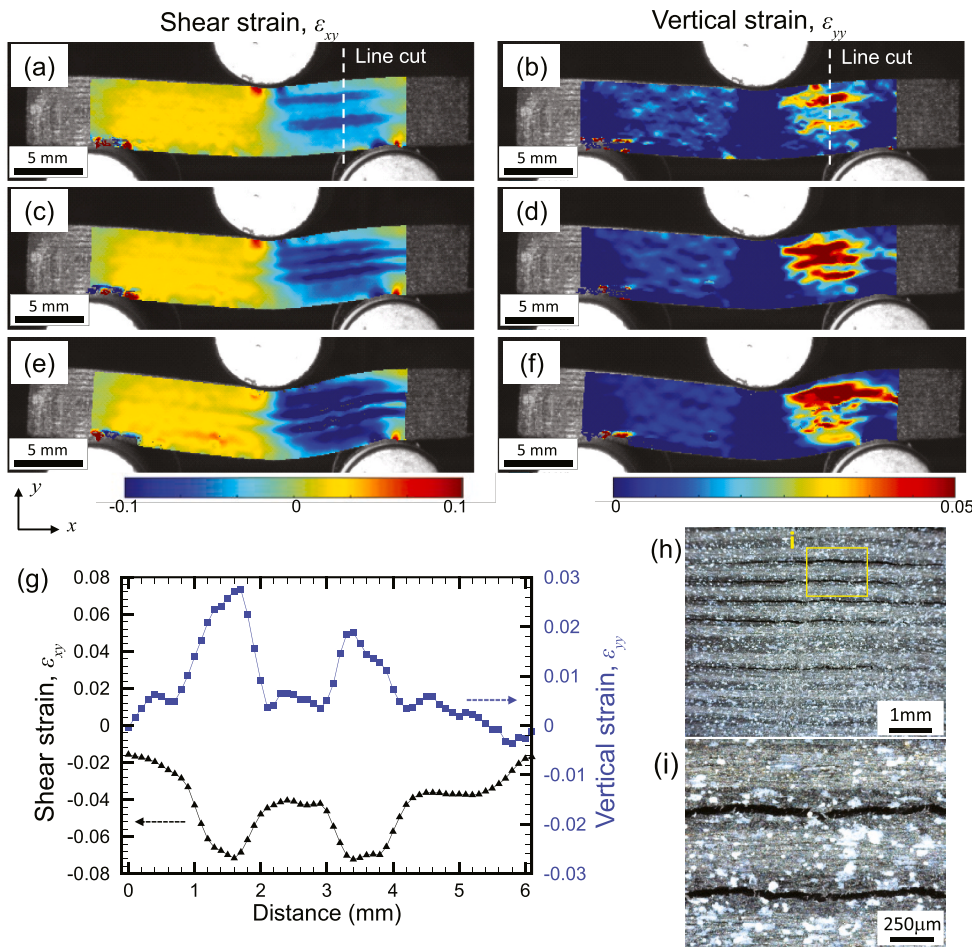


Fig. 10. The shear strain ϵ_{xy} and vertical strain ϵ_{yy} fields measured by the DIC method showing (a, b) the interlaminar damage nucleation state at point A marked in Fig. 9 (b) corresponding to the first stress drop, (c, d) damage evolution state at point B marked in Fig. 9(b) corresponding to the second stress drop, and (e, f) damage evolution state at point C marked in Fig. 9(b) corresponding to the third stress drop. (g) the ϵ_{xy} and ϵ_{yy} distributions along the vertical line cut shown in (a) and (b) as a function of the distance from the top surface, (h) an optical image showing the corresponding interlaminar cracking pattern on the sample side surface after the completion of the test, (i) a close-up view of the region i marked in (h) depicting bulk cracks within the short CFRP layers.

commences with the nucleation of the matrix crack in the top layers as shown in Fig. 13. Then, the maximum stress level remains increasing while the crack nucleation continues in the neighbor layers toward the bottom of the sample as shown in Fig. 13 (point B). Finally, the maximum shear stress curve shows a drop after reaching a critical density of matrix cracks in all short CFRP layers. Then the matrix cracks propagate within the short CFRP layers as shown in the snapshot at point C in Fig. 13. The numerically obtained maximum shear stress-displacement curve does not exhibit consecutive abrupt stress drops as observed in the experiments. This was associated with the progressive failure of different layers at each stress drop due to the stochastic distribution of defects in each short CFRP layer with a variable density. This results in a variable bulk fracture strength and toughness in each layer. However, the stochastic distribution of defects was not considered in the current numerical simulations. Therefore, the numerical curve shows only one major stress drop as a result of the uniform failure of the short CFRP layers. Overall, the numerically obtained maximum shear stress-displacement curve was able to reproduce the experimentally derived counterpart qualitatively and quantitatively. In addition, the matrix cracking pattern obtained from the simulations shown in Fig. 13 was found to be in a good agreement with the experimentally observed counterpart shown in Fig. 10(h).

Moreover, Fig. 14 shows the numerically derived maximum shear stress-displacement curve for the stacking sequence of $[(CF)_4, (SF)_4]_{6S}$, along with one of the experimentally measured counterparts. The inset snapshots in Fig. 14 reveal the three subsequent stages of the interfacial damage evolution and corresponding plastic strain fields at points A, B and C marked in the maximum shear stress-displacement curve. In the former, the color bar represents the damage parameter of the cohesive

surfaces between the continuous and short CFRP layers ranging from 0 to 1, where 1 denotes fully damaged cohesive elements. In the latter, the color bar shows the effective plastic strain with a threshold strain of 1%. Similar to the experimental counterpart, the numerically derived maximum shear stress-displacement curve initially shows a linear deformation. Upon reaching a critical stress state, both interfacial damage and plasticity commence simultaneously as shown in the snapshots captured at point A in Fig. 14. The interfacial damage nucleates at the second layer from the top surface. In addition, almost all the short CFRP layers reveal modest level of plasticity. From point A to B, the interfacial crack spreads to the subsequent layers towards the bottom interface, while the plasticity fully develops within the short CFRP layers. This results in a substantial nonlinearity observed from point A to B. Then, the maximum shear stress starts decreasing continuously as the interfacial cracks propagate along the interfaces between the continuous and short CFRP layers. Inspecting the interfacial damage pattern at point C in Fig. 14, one can notice that the 2nd and 4th interfaces above the mid-plane are damaged, while the 1st, 3rd, and 5th interfaces below the mid-plane are damaged. Such an interfacial cracking was found to be very consistent with the experimentally observed interfacial cracking pattern shown in Fig. 11(h). Moreover, the numerically obtained maximum shear stress-displacement curve was able to reproduce the experimentally derived counterpart.

5. Conclusions

In this study, we examined the interlaminar shear fracture behavior of 3D printed continuous and short CFRP composites for varying layouts and stacking sequences of the continuous and short CFRP layers. A

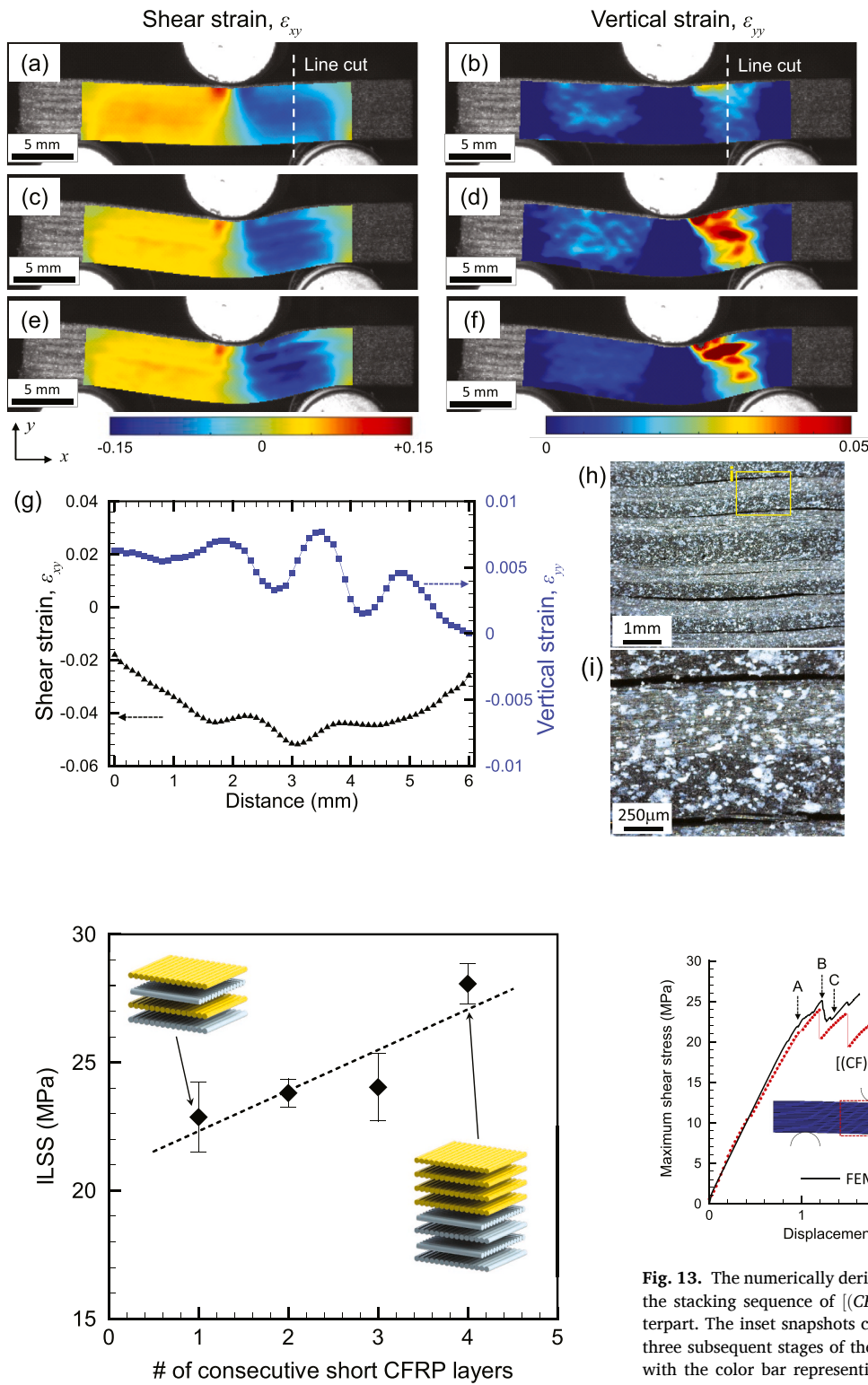


Fig. 12. The variation of the ILSS with the number of the consecutive short CFRP layers. The inset schematics show the stacking sequences for two different cases.

combined experimental and numerical study was carried out. In the experiments, SBS tests were performed to quantitatively measure the ILSS of the 3D printed continuous and short CFRP composites. The DIC technique was utilized to measure the in-situ full field strain fields and evaluate the interlaminar failure mechanisms from the shear and

Fig. 11. The shear strain ε_{xy} and vertical strain ε_{yy} fields measured by the DIC method showing (a, b) the interlaminar damage nucleation stage at point A marked in Fig. 9 (d) corresponding to the commencement of non-linearity, (c, d) damage evolution stage at point B marked in Fig. 9(d) corresponding to the peak value of the maximum shear stress, and (e, f) damage evolution stage at point C marked in Fig. 9(d) within the decaying portion of the curve. (g) the ε_{xy} and ε_{yy} distributions along the vertical line cut shown in (a) and (b) as a function of the distance from the top surface, (h) an optical image showing the corresponding interlaminar cracking pattern on the sample side surface after the completion of the test, (i) a close-up view of the region i marked in (h) depicting interfacial cracks at interfaces between the continuous and short CFRP layers.

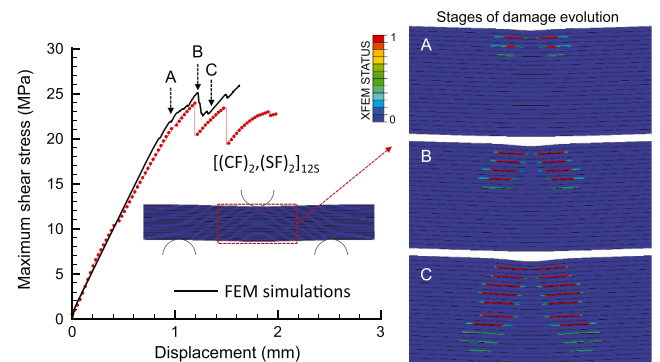


Fig. 13. The numerically derived maximum shear stress-displacement curve for the stacking sequence of $[(CF)_2, (SF)_2]_{12S}$, along with the experimental counterpart. The inset snapshots captured from the numerical simulation show the three subsequent stages of the damage evolution within the short CFRP layers with the color bar representing the status of the XFEM elements. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

vertical strain distributions on the side surfaces of the SBS samples. In the numerical study, FE models incorporating XFEM and CZM were developed to simulate the SBS tests. The result of the numerical simulation was used to understand the role of the stacking sequence of the continuous and short CFRP layers on the ILSS and resulting interlaminar failure modes. The following conclusions were drawn from this study:

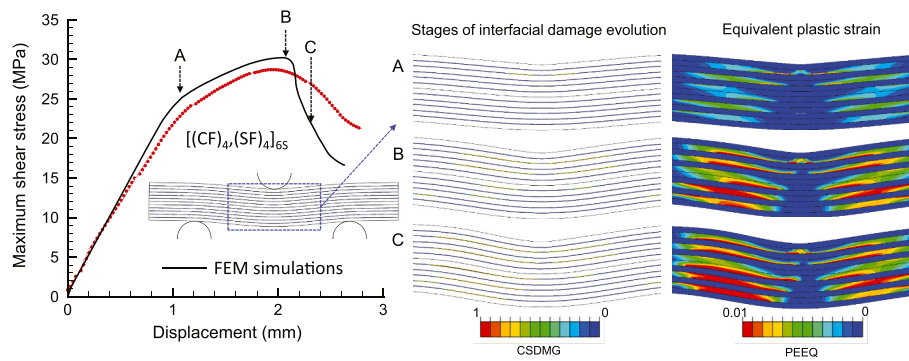


Fig. 14. The numerically derived maximum shear stress-displacement curve for the stacking sequence of $[(CF)_4,(SF)_4]_{6S}$, along with the experimental counterpart. The inset snapshots captured from the numerical simulation show the three subsequent stages of the interfacial damage evolution with the color bar representing the damage parameter of the cohesive surfaces between the continuous and short CFRP layers and corresponding equivalent plastic strain distributions within the short CFRP layers. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

1. The intrinsic ILSS of the 3D printed fully continuous CFRP composite was measured to be 40.9 ± 3 MPa, while the average ILSS of the 3D printed partially continuous and short CFRP composites was measured to be 24.4 ± 2.4 MPa. This suggested that the addition of short CFRP layers to replace 50% of the continuous CFRP layers leads to a significant reduction in the ILSS of the 3D printed fully continuous CFRP composites. The reduction in the ILSS can reach up to about 45%.
2. The ILSS of the 3D printed partially continuous and short CFRP composites was observed to be highly dependent on the stacking sequence of the continuous and short CFRP layers. About 25% deviation in the ILSS was measured with the varying stacking sequence of the continuous and short CFRP layers.
3. The ILSS of the 3D printed partially continuous and short CFRP composites exhibited a tendency to increase with the increasing number of consecutive short CFRP layers, which was accompanied by a failure mode transition from brittle matrix cracking of the short CFRP layers to the interfacial cracking of the interface between the continuous and short CFRP layers.
4. The combined experimental and numerical results showed that the increase in thickness of the short CFRP layers effectively enhances the fracture strength and toughness of the short CFRP layers, which prevents the short CFRP layers from brittle matrix cracking at a lower ILSS level and leads to the enhanced plastic deformation of the short CFRP layers. Therefore, the increasing thickness of the short CFRP layers results in greater ILSS of the partially continuous and short CFRP composites.

CRedit authorship contribution statement

Denizhan Yavas: Conceptualization, Methodology, Validation, Writing - original draft. **Ziyang Zhang:** Investigation. **Qingyang Liu:** Visualization. **Dazhong Wu:** Conceptualization, Methodology, Writing - review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] D.D. Chung, Composite materials: science and applications, Springer Science & Business Media 2010.
- [2] P.K. Mallick, Fiber-reinforced composites: materials, manufacturing, and design, CRC press 2007.
- [3] Ngo TD, Kashani A, Imbalzano G, Nguyen KT, Hui D. Additive manufacturing (3D printing): a review of materials, methods, applications and challenges. *Compos B Eng* 2018;143:172–96.
- [4] Kabir SF, Mathur K, Seyam A-FM. A critical review on 3D printed continuous fiber-reinforced composites: history, mechanism, materials and properties. *Compos Struct* 2020;232:111476.
- [5] Justo J, Távora L, García-Guzmán L, París F. Characterization of 3D printed long fibre reinforced composites. *Compos Struct* 2018;185:537–48.
- [6] Goh GD, Dikshit V, Nagalingam AP, Goh GL, Agarwala S, Sing SL, Wei J, Yeong WY. Characterization of mechanical properties and fracture mode of additively manufactured carbon fiber and glass fiber reinforced thermoplastics. *Mater Des* 2018;137:79–89.
- [7] Al Abadi H, Thai H-T, Paton-Cole V, Patel V. Elastic properties of 3D printed fibre-reinforced structures. *Compos Struct* 2018;193:8–18.
- [8] Peng Y, Wu Y, Wang K, Gao G, Ahzi S. Synergistic reinforcement of polyamide-based composites by combination of short and continuous carbon fibers via fused filament fabrication. *Compos Struct* 2019;207:232–9.
- [9] Ning F, Cong W, Hu Y, Wang H. Additive manufacturing of carbon fiber-reinforced plastic composites using fused deposition modeling: effects of process parameters on tensile properties. *J Compos Mater* 2017;51(4):451–62.
- [10] Ning F, Cong W, Qiu J, Wei J, Wang S. Additive manufacturing of carbon fiber reinforced thermoplastic composites using fused deposition modeling. *Compos B Eng* 2015;80:369–78.
- [11] Tekinalp HL, Kunc V, Velez-Garcia GM, Duty CE, Love LJ, Naskar AK, Blue CA, Ozcan S. Highly oriented carbon fiber-polymer composites via additive manufacturing. *Compos Sci Technol* 2014;105:144–50.
- [12] Karsli NG, Aytaç A. Tensile and thermomechanical properties of short carbon fiber reinforced polyamide 6 composites. *Compos B Eng* 2013;51:270–5.
- [13] Zhong W, Li F, Zhang Z, Song L, Li Z. Short fiber reinforced composites for fused deposition modeling. *Mater Sci Eng, A* 2001;301(2):125–30.
- [14] Chacón J, Caminero M, Núñez P, García-Plaza E, García-Moreno I, Reverte J. Additive manufacturing of continuous fibre reinforced thermoplastic composites using fused deposition modelling: effect of process parameters on mechanical properties. *Compos Sci Technol* 2019;181:107688.
- [15] Yu T, Zhang Z, Song S, Bai Y, Wu D. Tensile and flexural behaviors of additively manufactured continuous carbon fiber-reinforced polymer composites. *Compos Struct* 2019;225:111147.
- [16] Caminero M, Chacón J, García-Moreno I, Reverte J. Interlaminar bonding performance of 3D printed continuous fibre reinforced thermoplastic composites using fused deposition modelling. *Polym Test* 2018;68:415–23.
- [17] Caminero MA, Chacón JM, García-Plaza E, Núñez PJ, Reverte JM, Becar JP. Additive manufacturing of PLA-based composites using fused filament fabrication: effect of graphene nanoplatelet reinforcement on mechanical properties, dimensional accuracy and texture. *Polymers* 2019;11(5):799.
- [18] ISO/ASTM52921-13 (2019). Standard Terminology for Additive Manufacturing—coordinate systems and test methodologies. West Conshohocken, PA: ASTM International; 2019.
- [19] ASTM D2344/D2344M-16. Standard test method for short-beam strength of polymer matrix composite materials and their laminates. West Conshohocken, PA: ASTM International; 2016.
- [20] Blaber J, Adair B, Antoniou A. Ncorr: open-source 2D digital image correlation matlab software. *Exp Mech* 2015;55(6):1105–22.
- [21] Belytschko T, Black T. Elastic crack growth in finite elements with minimal remeshing. *Int J Numer Methods Eng* 1999;45(5):601–20.
- [22] Bouhala L, Makradi A, Belouettar S, Kiefer-Kamal H, Frères P. Modelling of failure in long fibres reinforced composites by X-FEM and cohesive zone model. *Compos B Eng* 2013;55:352–61.
- [23] Ye C, Shi J, Cheng GJ. An extended finite element method (XFEM) study on the effect of reinforcing particles on the crack propagation behavior in a metal-matrix composite. *Int J Fatig* 2012;44:151–6.
- [24] Viguera G, Sket F, Samaniego C, Wu L, Noels L, Tjahjanto D, Casoni E, Houzeaux G, Makradi A, Molina-Aldareguia JM. An XFEM/CZM implementation for massively parallel simulations of composites fracture. *Compos Struct* 2015;125:542–57.
- [25] Zhao L, Wang Y, Zhang J, Gong Y, Hu N, Li N. XFEM-based model for simulating zigzag delamination growth in laminated composites under mode I loading. *Compos Struct* 2017;160:1155–62.
- [26] Kao Y-T, Zhang Y, Wang J, Tai BL. Bending behaviors of 3D-printed Bi-material structure: experimental study and finite element analysis. *Additive Manufacturing* 2017;16:197–205.

- [27] Li S, Thouless M, Waas A, Schroeder J, Zavattieri P. Use of mode-I cohesive-zone models to describe the fracture of an adhesively-bonded polymer-matrix composite. *Compos Sci Technol* 2005;65(2):281–93.
- [28] Li S, Thouless M, Waas A, Schroeder J, Zavattieri P. Use of a cohesive-zone model to analyze the fracture of a fiber-reinforced polymer-matrix composite. *Compos Sci Technol* 2005;65(3–4):537–49.
- [29] Farmand-Ashtiani E, Alanis D, Cugnoni J, Botsis J. Delamination in cross-ply laminates: identification of traction-separation relations and cohesive zone modeling. *Compos Sci Technol* 2015;119:85–92.
- [30] Yavas D, Shang X, Bastawros AF. Mode-I fracture toughness and surface morphology evolution for contaminated adhesively bonded composite structures. *Compos Struct* 2018;203:513–22.
- [31] Geubelle PH, Baylor JS. Impact-induced delamination of composites: a 2D simulation. *Compos B Eng* 1998;29(5):589–602.
- [32] Ghandriz R, Hart K, Li J. Extended finite element method (XFEM) modeling of fracture in additively manufactured polymers. *Additive Manufacturing* 2020;31: 100945.
- [33] Tran P, Ngo TD, Ghazlan A, Hui D. Bimaterial 3D printing and numerical analysis of bio-inspired composite structures under in-plane and transverse loadings. *Compos B Eng* 2017;108:210–23.
- [34] Smith M. ABAQUS/Standard user's manual, version 6.13. Providence, RI: Dassault Systèmes Simulia Corp; 2013.
- [35] https://static.markforged.com/markforged_composites_datasheet.pdf.
- [36] Turon A, Davila CG, Camanho PP, Costa J. An engineering solution for mesh size effects in the simulation of delamination using cohesive zone models. *Eng Fract Mech* 2007;74(10):1665–82.
- [37] Hillerborg A, Modéer M, Petersson P-E. Analysis of crack formation and crack growth in concrete by means of fracture mechanics and finite elements. *Cement Concr Res* 1976;6(6):773–81.
- [38] He Q, Wang H, Fu K, Ye L. 3D printed continuous CF/PA6 composites: effect of microscopic voids on mechanical performance. *Compos Sci Technol* 2020;191: 108077.
- [39] Chabaud G, Castro M, Denoual C, Le Duigou A. Hygromechanical properties of 3D printed continuous carbon and glass fibre reinforced polyamide composite for outdoor structural applications. *Additive Manufacturing* 2019;26:94–105.
- [40] Blok LG, Longana ML, Yu H, Woods BK. An investigation into 3D printing of fibre reinforced thermoplastic composites. *Additive Manufacturing* 2018;22:176–86.
- [41] van de Werken N, Hurley J, Khanbolouki P, Sarvestani AN, Tamijani AY, Tehrani M. Design considerations and modeling of fiber reinforced 3D printed parts. *Compos B Eng* 2019;160:684–92.
- [42] T.L. Anderson, *Fracture mechanics: fundamentals and applications*, CRC press 2017.
- [43] Chai H. Observation of deformation and damage at the tip of cracks in adhesive bonds loaded in shear and assessment of a criterion for fracture. *Int J Fract* 1993;60(4):311–26.
- [44] Groleau M, Shi Y-B, Yee A, Bertram J, Sue H, Yang P. Mode II fracture of composites interlayered with nylon particles. *Compos Sci Technol* 1996;56(11): 1223–40.
- [45] Kin Loch A, Shaw S. The fracture resistance of a toughened epoxy adhesive. *J Adhes* 1981;12(1):59–77.